

Assignment - Series 01 RStudio

Martin Pigeau

13 September 2026, 14:21:08

1. Conclusion in 4.1

My conclusion is the same as his. The Wilcoxon-Mann-Whitney test I performed showed a statistically significant difference in weight between male and female students ($p = 0.003$), with a large effect size ($r = 0.63$). These values are very close to the ones given in the assignment ($p = 0.0031$, $r = 0.6$), leading to the same overall conclusion: male students in this sample weigh significantly more than female students, and this difference is not only statistically significant but also practically meaningful given the large effect size.

2. Three most important things learnt

1. **Building real, reusable comparative graphs.** I learned to construct plots such as boxplots, histograms, and confidence intervals that are specifically designed to answer a comparative question, and that can be reused as a template for future projects and studies.
2. **Applying statistical tests learned in my Bachelor's directly in RStudio.** I put into practice concepts I had only seen theoretically before, such as normality testing, non-parametric tests, and effect size. I also learned how to choose the right test based on the actual properties of the data rather than applying one by default.
3. **Learning to use a LLM as a learning aid rather than a shortcut.** Throughout this series, I learned to ask targeted questions, verify explanations against official R documentation, and critically check the answers I received rather than copying them blindly.